S.S.V.P.S's B. S. Deore Polytechnic, Dhule

Department of Computer Engineering

Academic Year : 2025 – 2026

Presentation on Capstone Project Execution (CPE–316004)

Topic	**The Baby Vaccination Management System**	
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Abstract

- The **Baby Vaccination Management System** is a web application developed to help manage and monitor baby vaccination schedules. The system keeps digital records of baby details, vaccination dates, and vaccination history. It helps parents get reminders for upcoming vaccines and ensures that babies receive vaccinations on time. Doctors and healthcare staff can easily update and manage vaccination records using this system. This digital system helps hospitals reduce paperwork and improve child healthcare management.

Content

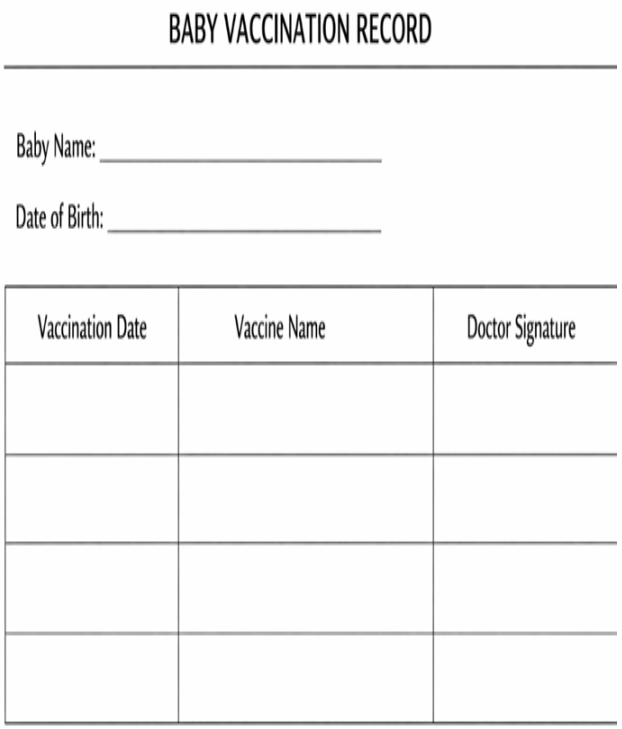
1. Introduction (background of the industry or used based problem / task)
2. Literature Survey
3. Scope of the project
4. Methodology
5. Details of designs, working and processes
6. Results and applications
7. Conclusion and future scope
8. References and Bibliography

1. Introduction

- Vaccination is very important to protect babies from serious diseases.
- Hospitals usually keep vaccination records manually.
- Manual records can cause errors and loss of important data.
- Parents sometimes forget vaccination dates.
- The **Baby Vaccination Management System** helps manage vaccination records digitally.
- It allows parents and doctors to track vaccination schedules easily.
- The system helps improve child healthcare management.

2. Literature Review

- Many hospitals still keep **vaccination records manually**. This can cause mistakes and loss of data.
- Some healthcare systems store **patient data digitally**, but they do not focus only on baby vaccination.
- Digital systems help hospitals **store data safely and manage records easily**.
- Reminder systems help parents **remember vaccination dates** and avoid missing vaccines.
- The **Baby Vaccination Management System** helps parents and doctors **manage baby vaccination records digitally**.
- This system helps **reduce paperwork, save time, and improve child healthcare management**.



BABY VACCINATION RECORD

Baby Name: _____

Date of Birth: _____

Vaccination Date	Vaccine Name	Doctor Signature

3. Scope of the project

- **Functional Scope**
  - Baby registration and profile management
  - Vaccination schedule generation
  - Vaccination status update
  - Viewing vaccination history
- **Limitations**
  - The system supports only basic vaccination management
  - Mobile application support is not available
- **Assumptions**
  - Users have basic knowledge of using web applications
  - Internet connection is available to access the system

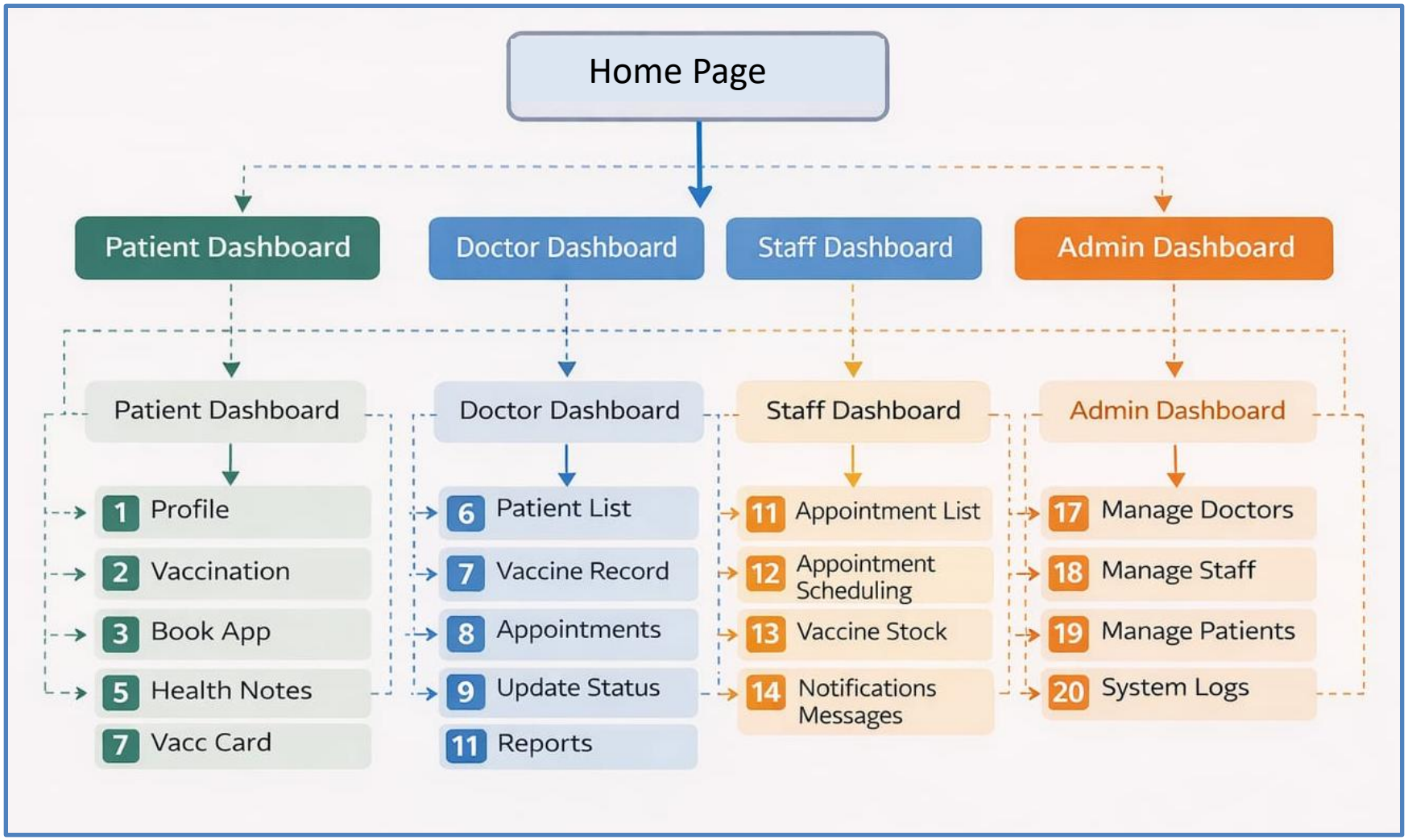
4. Methodology

- First, baby details are registered in the system by the user or hospital staff.
- The system stores baby information and generates a vaccination schedule based on age.
- Doctors or healthcare staff update vaccination status after each vaccination.
- The system maintains digital records of vaccination history.
- Parents can view vaccination schedules and track completed or pending vaccines.

4. Methodology

Patient	Doctor	Staff	Admin
1. Patient Registration			17. Manage Doctors
2. Login to System			18. Manage Staff
3. View Baby Profile			19. Manage Patients
4. View Vaccination Schedule	6. View Appointment Request	6. View Appointment Request	20. Manage Vaccine Stock
5. Book Appointment →	7. Accept / Reject Appointment	7. Accept / Reject Appointment	21. View Reports
↓	**8. Schedule Vaccination** →	**8. Schedule Vaccination** →	22. Monitor System
	9. Check Patient Details	9. Check Patient Details	
13. Patient views Vaccination Card	10. **Update Vaccination Status**	10. **Update Vaccination Status** →	Progress ↓
14. View Vaccine Status	11. Add Vaccine Record	11. Add Vaccine Record	
15. Receive Notification	**12. Add Health Notes**	**12. Add Health Notes**	
16. View Doctor Notes	↓		

Web-Application Flow



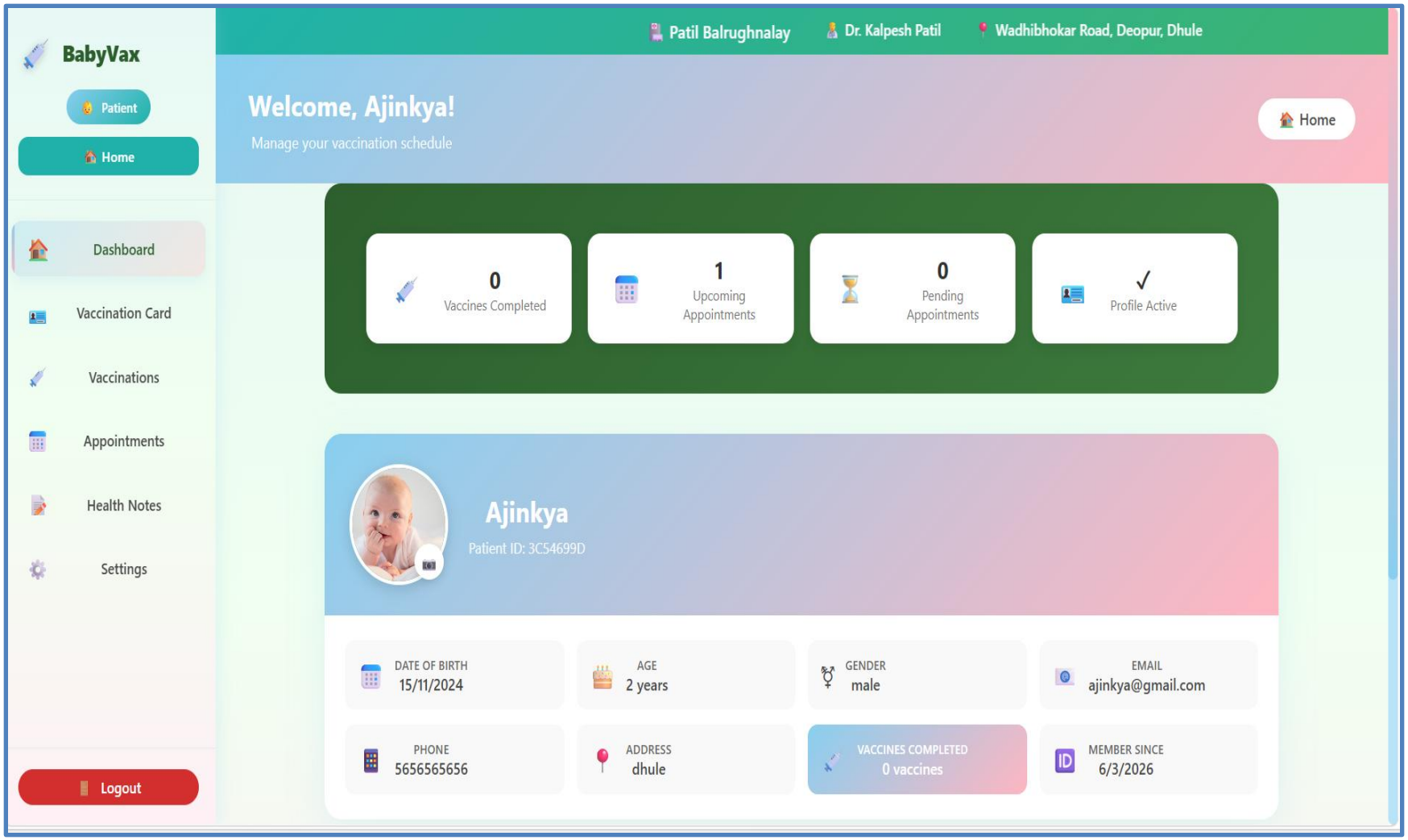
5. Details of designs, working & processes

- The Baby Vaccination Management System helps manage and track baby vaccination records digitally. It allows parents, doctors, and hospital staff to access vaccination information, update schedules, and store records securely in the database
- **System Working**
  - User/Admin login to the system
  - Baby registration in the system
  - Vaccination schedule generation
  - Vaccination status update after vaccination
  - Data stored securely in the database
  - Users can view vaccination history through dashboard

5. Details of designs, working & processes

- **System Users**
  - **Parent** – Register baby and view vaccination schedule
  - **Doctor** – Update vaccination status and check records
  - **Admin** – Manage system, users, and reports
  - **Hospital Staff** – Register baby details and assist in vaccination record

6. Results and Applications



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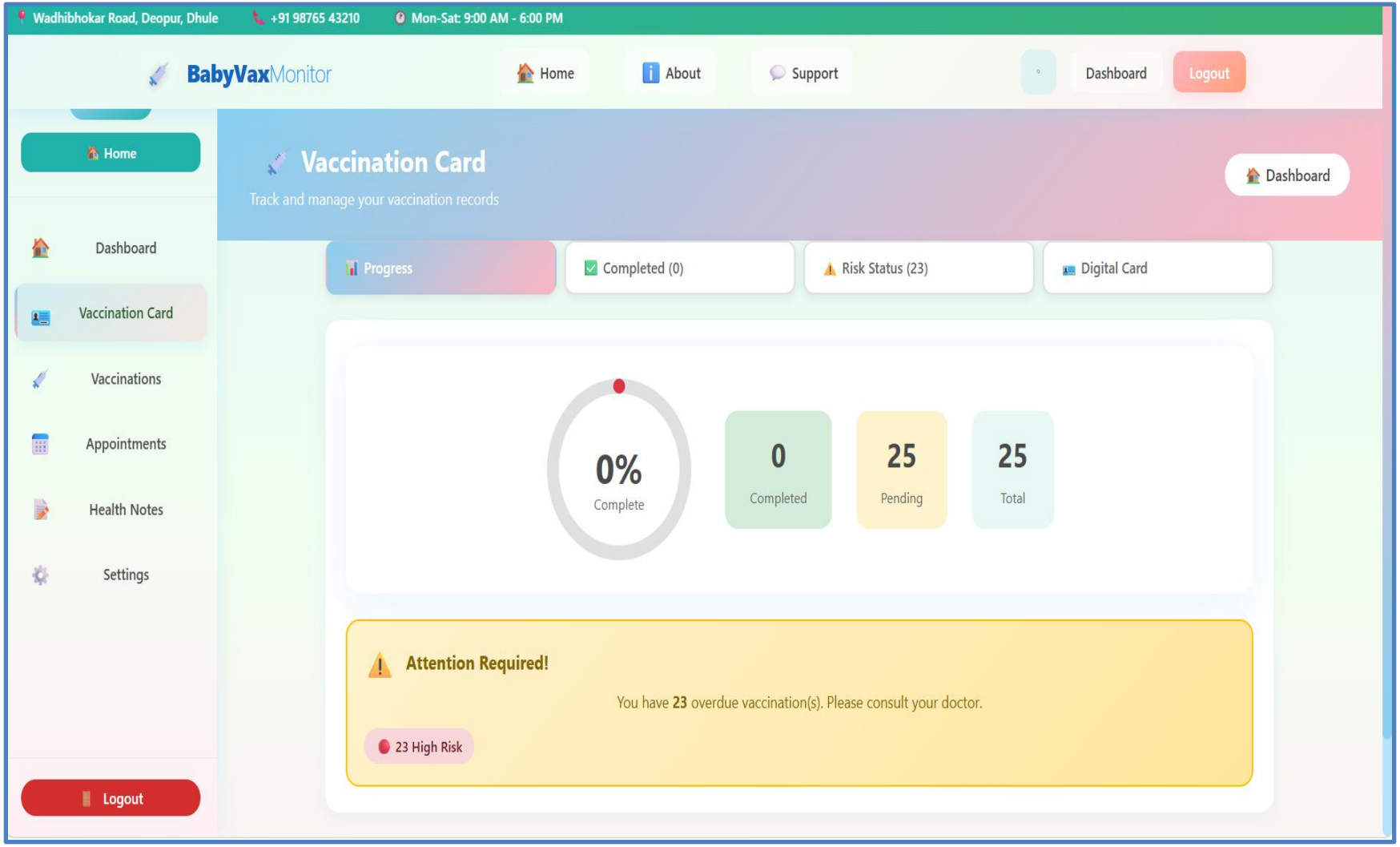

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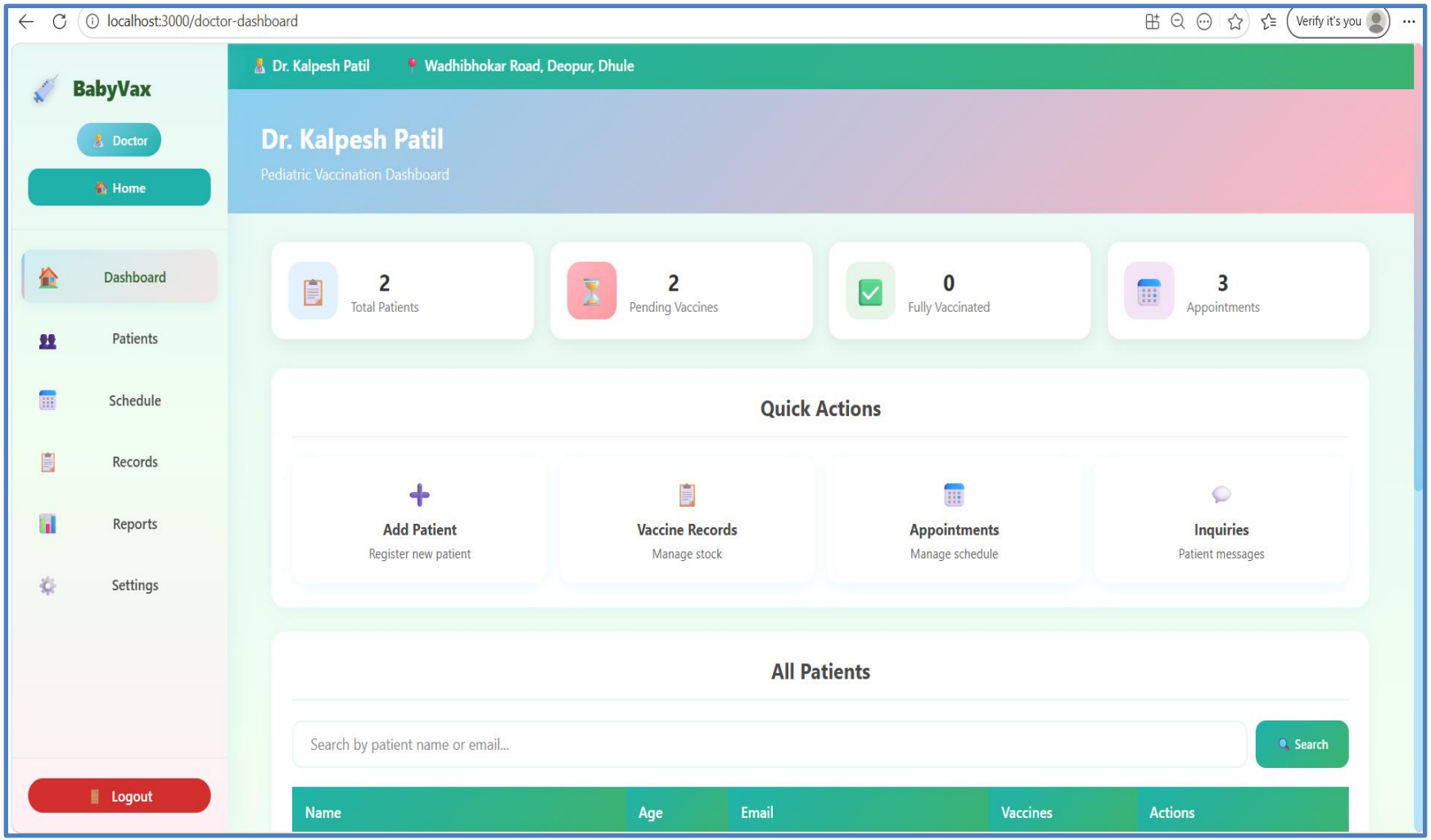

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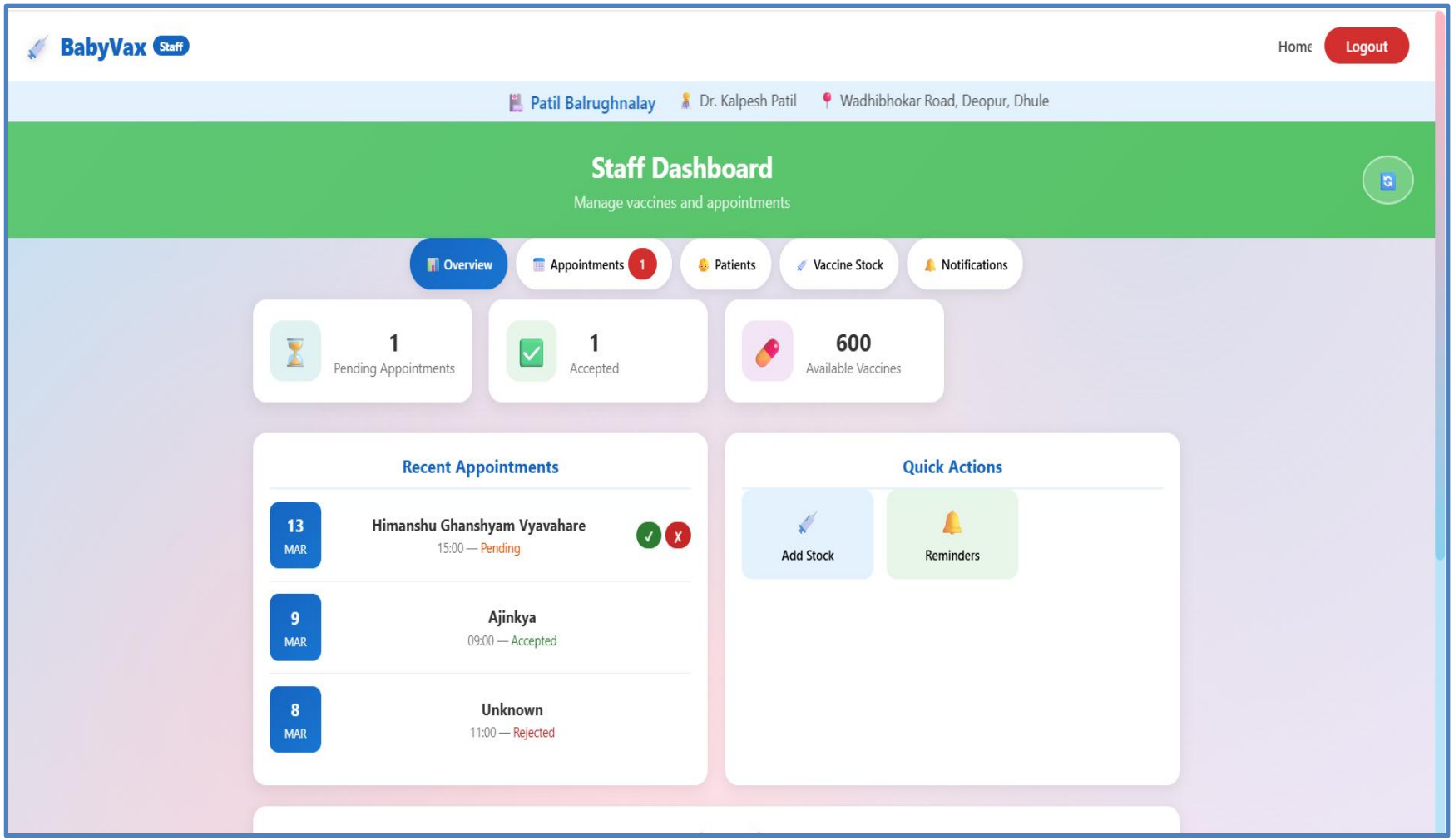

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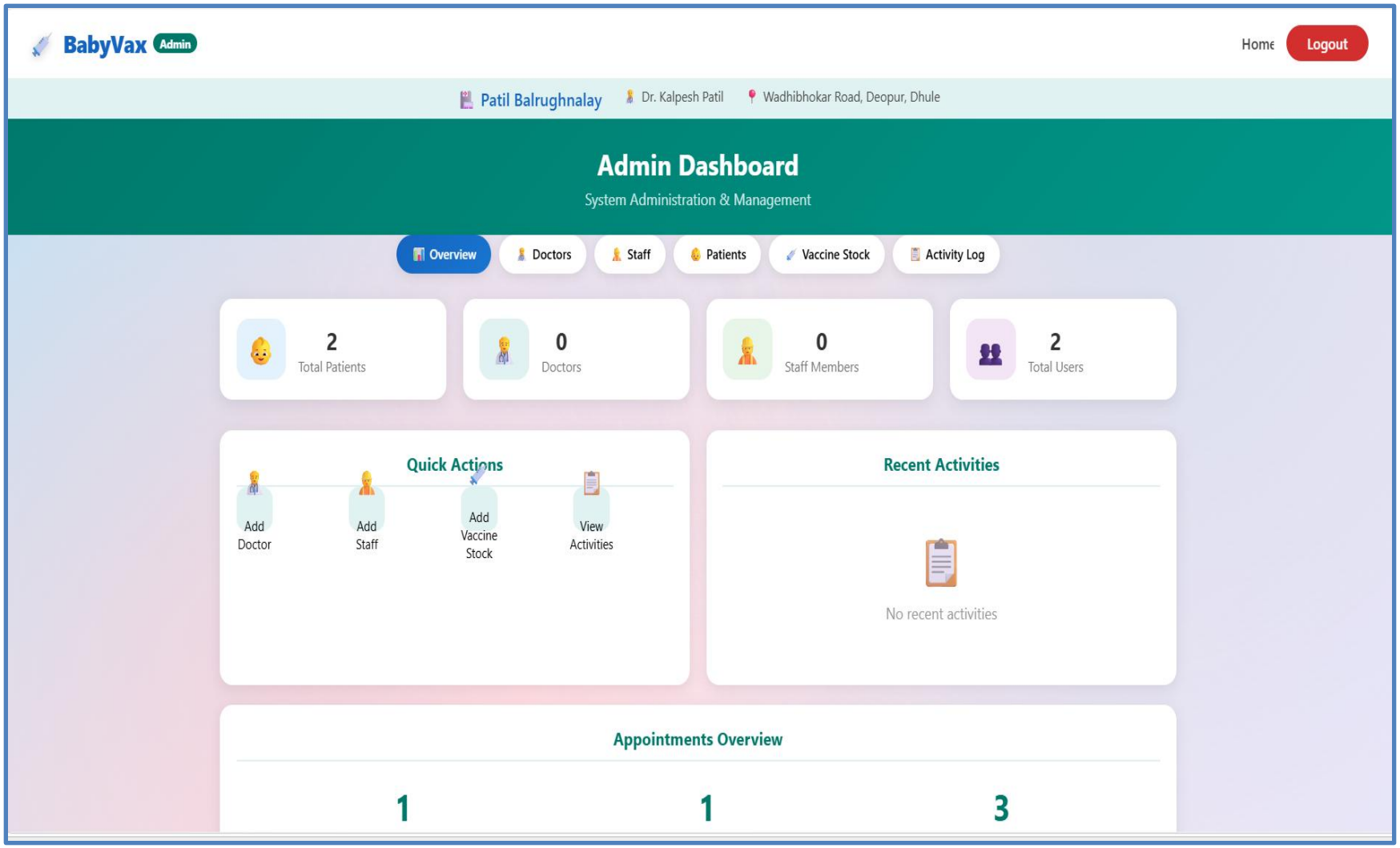

7. Conclusion and future scope

- The Baby Vaccination Management System is proposed to provide an efficient and reliable solution for managing infant vaccination records. The system aims to replace manual record-keeping with a digital platform that improves accuracy, accessibility, and efficiency. By ensuring timely vaccination and secure record management, the proposed system benefits parents, healthcare providers, and medical institutions. Overall, the project contributes to improved child healthcare management and digital transformation in the healthcare sector
- **Future Scope**
  - SMS and Email notification system for vaccination reminders
  - Mobile application support
  - Multi-language support
  - Cloud-based deployment
  - Advanced reporting and analytics

8. References and Bibliography

- World Health Organization – Child Vaccination Information - https://www.who.int
- Ministry of Health and Family Welfare (India) https://www.mohfw.gov.in
- Dr. Kalpesh Patil, MD (Balrog Specialist / Pediatrician)